

Cameron Petrie

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Portfolio: <https://campetrie.com>

GitHub: <https://github.com/cam-petrie>

EDUCATION

University of Colorado, Boulder, CO, United States

Master of Science in Computer Science (Currently Pursuing)

Cumulative GPA: 3.9/4.0

Miami University, Oxford, OH, United States

Bachelor of Arts in Economics

Bachelor of Arts in Political Science, Public Administration

TECHNICAL SKILLS

Backend Development: Java, Python, Node.js, Fastify, Express.js, Drizzle ORM, Mongoose, PostgreSQL, MongoDB, Neo4j, Kotlin, REST APIs, JSON/XML

Frontend Development: React, Redux, Zustand, TypeScript/JavaScript, Next.js, Vite, Tailwind CSS, HTML, CSS, jQuery, Angular, Bootstrap, D3/SVG data visualization, i18n

Deployment & Version Control: CI/CD, Docker, Kubernetes (GKE), Git, GitHub

Miscellaneous: Generative AI & LLMs, Causal Inference (LiNGAM / linear-SEM), Organizational Network Analysis, Machine Learning, Neo4j graph modeling, Bull job queues, Mailgun, GraphQL, Microservices, Distributed Systems, Software Architecture, UNIX/Linux, Agile (JIRA, Confluence), Playwright, JUnit, OOP

EXPERIENCE

Cognitive Talent Solutions

Feb 2025 – Current

Chief Technical Officer

- Own the technical vision and architecture for an AI-driven Organizational Network Analysis (ONA) platform: a Fastify/PostgreSQL backend, Neo4j graph store, and React/TypeScript frontend on Kubernetes (GKE).
- Lead the “Causal AI” initiative — an interactive dashboard rendering multi-node causal graphs (LiNGAM / linear-SEM) with signed effect sizes, significance, and bootstrap stability to guide employee-retention decisions.
- Introduced a generative-AI assistant that explains fitted causal models in plain language for non-technical stakeholders, and defined the team’s engineering, code-review, and CI/CD standards.

Cognitive Talent Solutions

Feb 2024 – Feb 2025

Principal Software Engineer

- Architected core survey/ONA services (Fastify, Drizzle ORM, PostgreSQL, Neo4j) with Bull/Mailgun email pipelines driving large-scale, multi-language participant campaigns.
- Diagnosed and fixed a systemic data-integrity defect in which survey reminder counts diverged from the true recipient population, correcting the metric across thousands of surveys.
- Built a generative-AI chatbot and an AI data-onboarding tool (header standardization, language detection), plus an interactive results dashboard with styled PDF export and Playwright E2E tests.

PROJECTS

Project: Network-First Manifesto DAO — Web3 Governance & ONA

2025 – 2026

- Launched a Web3 DAO (Next.js 14, Privy embedded wallets, Base L2) with a Snapshot voting + Gnosis Safe treasury stack for CTS’s Network-First Manifesto.
- Built a DAO-specific Organizational Network Analysis methodology — voting patterns, governance participation, network topology — to productize commercial ONA services for DAO communities.

Project: Causal AI Network — Retention Decision Intelligence

2025

- Built an interactive causal-network dashboard visualizing multi-node LiNGAM / linear-SEM models as a directed graph — signed standardized β , p-values, and bootstrap stability.
- Layered a generative-AI assistant (“Cogni”) that narrates the strongest drivers and recommended actions; shipped as an embedded React feature and a self-contained offline demo.

Project: AI-Powered Organizational Network Analysis Platform

2024 – 2025

- Full-stack ONA platform (Fastify, Drizzle ORM, PostgreSQL, Neo4j, React/TypeScript, Zustand) modeling employees and their working relationships as an interactive network graph.
- Implemented the full survey lifecycle — onboarding, edge collection, completion tracking, and a Bull/Mailgun reminder engine — deployed to GKE with Docker and CI/CD.